

MARSHALL, AK, USA

WMO: 702628

Lat: 61.867N

Lon: 162.033W

Elev: 31

StdP: 100.95

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00105

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-29.2	-35.1	0.1	-30.9	-32.8	0.2	-28.8	19.3	-6.2	18.2	-7.3	1.4	70	0.837

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	23.7	15.2	21.6	14.4	19.0	13.5	16.4	21.8	15.2	20.0	14.1	18.0	4.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	17.8	13.0	9.3	16.6	12.2	8.9	15.8	45.7	21.7	42.3	20.2	39.5	18.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.9	11.9	DB	-37.1	27.4	4.0	3.0	-40.0	29.6	-42.3	31.3	-44.6	33.0	-47.5	35.1	(4)
(5)			WB	-34.7	17.2	2.4	1.6	-36.5	18.3	-37.9	19.3	-39.2	20.2	-41.0	21.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-12.2	-10.2	-9.2	-1.4	5.9	11.2	12.8	11.5	7.5	0.6	-8.9	-11.5	(6)	
	DBStd	11.37	10.14	8.67	7.55	6.38	4.54	3.15	2.90	2.84	3.62	4.89	7.56	8.62	(7)	
	HDD10.0	3986	687	564	595	343	139	21	7	16	87	293	567	667	(8)	
	HDD18.3	6796	945	798	853	593	387	217	174	212	324	551	817	925	(9)	
	CDD10.0	236	0	0	0	0	11	56	93	63	13	1	0	0	(10)	
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0	(11)	
	CDH23.3	81	0	0	0	0	2	25	49	5	0	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	5	8	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.0	7.2	6.8	5.3	4.5	4.1	3.7	3.9	3.8	4.3	5.1	5.3	6.6	(14)	
(15) Precipitation	PrecAvg	618	43	38	34	32	32	46	68	80	75	62	60	49	(15)	
	PrecMax	767	93	112	125	68	73	80	120	142	142	130	138	108	(16)	
	PrecMin	466	12	7	8	9	10	16	25	28	19	10	3	20	(17)	
	PrecStd	84	21	25	24	15	15	13	21	24	29	28	33	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	4.8	7.0	13.8	22.8	26.1	27.8	23.8	17.9	12.0	5.9	7.5	(19)	
		MCWB	3.8	2.7	3.5	6.8	13.0	15.7	17.8	16.0	11.9	8.3	4.1	3.0	(20)	
	2%	DB	4.1	3.1	3.8	11.4	18.5	23.1	24.4	21.0	16.0	10.0	4.3	3.8	(21)	
		MCWB	2.1	1.2	1.3	5.7	10.7	14.2	16.3	14.8	11.1	7.5	3.1	1.8	(22)	
	5%	DB	2.7	2.1	2.3	9.1	16.2	20.9	22.0	18.7	14.0	8.6	2.8	1.9	(23)	
		MCWB	1.4	0.6	0.5	4.6	9.5	13.4	15.6	13.7	10.6	7.0	2.0	0.7	(24)	
	10%	DB	1.0	1.0	1.0	7.3	13.8	18.3	19.1	17.3	12.7	7.4	1.2	0.3	(25)	
		MCWB	0.0	-0.1	-0.5	3.5	8.3	12.3	14.3	12.9	10.3	6.0	0.5	-0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	2.9	3.7	7.5	13.6	16.3	19.6	16.9	13.1	9.2	4.6	5.0	(27)	
		MCDB	6.2	4.6	6.6	13.0	21.1	23.9	26.2	21.7	15.1	10.9	5.5	6.6	(28)	
	2%	WB	2.4	1.5	1.7	5.9	11.6	15.2	17.3	15.6	12.0	8.0	3.1	1.9	(29)	
		MCDB	3.9	2.7	3.2	10.8	18.2	21.8	23.3	19.5	14.9	9.4	3.9	3.3	(30)	
	5%	WB	1.3	0.7	0.6	4.8	9.9	13.8	15.8	14.5	11.1	7.2	2.0	0.8	(31)	
		MCDB	2.7	1.8	2.0	9.0	15.4	19.4	20.8	17.8	13.5	8.4	2.9	1.9	(32)	
	10%	WB	0.0	-0.1	-0.5	3.7	8.7	12.7	14.6	13.5	10.4	6.0	0.5	-0.5	(33)	
		MCDB	1.0	1.0	0.7	7.0	13.3	18.0	18.8	16.4	12.5	7.4	1.2	0.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.4	8.6	9.1	10.2	10.2	8.6	8.1	7.5	5.8	6.1	6.4	(35)	
		MCDBR	5.2	5.5	7.5	11.1	14.9	16.0	15.4	12.5	9.5	5.9	4.4	5.8	(36)	
	5% WB	MCWBR	4.6	4.7	5.8	6.9	8.2	8.2	8.6	7.0	5.9	4.5	4.1	5.0	(37)	
		MCWBR	5.5	5.8	6.7	10.6	13.6	13.8	13.3	10.0	8.2	5.3	4.4	5.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.217	0.275	0.280	0.351	0.373	0.365	0.367	0.384	0.331	0.291	0.230	0.200	(40)	
		taud	2.038	2.156	2.230	2.087	2.279	2.386	2.421	2.353	2.486	2.400	2.042	1.887	(41)	
	Edn at Noon	Edn at Noon	483	685	838	826	844	860	848	794	784	680	441	281	(42)	
		Edh at Noon	36	67	90	128	116	106	100	99	72	53	35	23	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.32	1.01	2.87	4.52	5.16	5.32	4.34	3.43	2.30	1.11	0.37	0.15	(44)	
	RadStd	RadStd	0.04	0.13	0.28	0.37	0.49	0.53	0.38	0.41	0.23	0.09	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page